

ABES Institute of Technology, Ghaziabad

CRC Training Program - 2026-27 Session Odd

Training Name - MongoDB and Introduction to NodeJS

B.Tech 5th Sem (2024-28 Batch)

	Prerequisite : JavaScript	
Table of Contents	Detailed Contents	Duration
JavaScript Refresher	ES6+ Intro, Let and Const (Replacing var)	4
	Functions-named, callback	
	Function expression, Arrow function	
	Array functions: forEach, filter, reduce, map	
	Template String, Spread/Rest	
	Destructuring, Promise	
Introduction to NoSQL & MongoDB	Introduction to Databases, SQL vs NoSQL, Types of NoSQL Databases, Features & Advantages of MongoDB, MongoDB Architecture, BSON & JSON, Installing MongoDB, MongoDB Compass, MongoDB Atlas, Basic Database Operations	2
MongoDB CRUD Operations	Creating Databases & Collections, Insert, Find, Update & Delete Documents, Query Operators, Comparison Operators, Logical Operators, Projection, Sorting, Limiting Results	6
Advanced MongoDB Concepts	Embedded Documents, Arrays, Indexing, Aggregation Pipeline, Group, Match, Project, Lookup, Data Modeling, Validation, Import & Export Data	4
Introduction to NodeJS	What is Node.js, JavaScript Runtime Environment, V8 Engine, Installing Node.js, npm & npx, REPL, Running Node Programs, Package Management	2
Core NodeJS Modules	Modules, CommonJS & ES Modules, File System Module, Path Module, OS Module, Events Module, Buffers, Streams, Process Object, Timers	4
Asynchronous Programming	Callback Functions, Promises, Async/Await, Event Loop, Non-blocking I/O, Error Handling, Working with APIs, HTTP Module	6
Express.js Basics	Introduction to Express.js, Express Installation, Creating Server, Routing, Middleware, Request & Response Objects, Serving Static Files, REST API Basics	4
MongoDB with NodeJS	MongoDB Driver, Connecting Node.js to MongoDB, CRUD Operations using Node.js, Environment Variables, Configuration Files, Project Structure	4
Project	Develop a CRUD-based REST API using Node.js, Express.js, and MongoDB	4
	Testing APIs using Postman, Project Deployment Basics	
Total		40